Thambawita Maddumage Sajith Priyankara

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Research Interests

Functional data analysis, object-oriented data analysis, statistical analysis on manifolds,high-dimensional statistical modeling, protein–ligand binding analysis, statistical methods for spatial transcriptomics and genomics, longitudinal and epidemiologic data analysis.

Education

Ph.D. Candidate in Statistics 2021–Present
Texas Tech University, Lubbock, TX

- Dissertation research focuses on statistical representations and predictive modeling of protein–ligand binding sites using functional data analysis, multivariate statistics, and geometry-aware methods.

M.S. in Statistics 2021–2023
Texas Tech University, Lubbock, TX

B.Sc. in Statistics 2015–2019
University of Peradeniya, Sri Lanka

Research Experience

Ph.D. Researcher : Statistical Modeling of Protein–Ligand Binding Sites 2021–Present
Department of Mathematics & Statistics, Texas Tech University

- Develop statistical representations of protein binding sites using functional data analysis to characterize structural and elemental variation in protein–ligand binding sites.
- Apply geometry-aware methods based on constrained functions, covariance structures, and statistical analysis on nonlinear spaces for comparing protein binding sites.
- Developed predictive models using statistical learning, machine learning, and deep learning approaches for ligand-binding classification and protein pocket matching, including random forests, penalized regression, and Siamese neural networks.
- Developed **FUSED**, a functional representation for joint structural and elemental analysis of protein-ligand binding sites; first-author manuscript **in revision for resubmission to the *Journal of Chemical Information and Modeling*** and available as a **bioRxiv preprint**.
- First-author manuscript, *Element-Specific Axis Distance Density (ESADD) Representations of Protein Binding Sites*, currently in final preparation.

Collaborative Biostatistics Researcher 2026–Present
Dr. Dawei Li Lab, Texas Tech University Health Sciences Center

- Collaborate with biomedical researchers on statistical analyses of alcohol use disorder, psychiatric comorbidities, early-life exposures, and related population health outcomes.
- Develop multivariable logistic regression, mixed-effects models, interaction analyses, multiple-imputation workflows, and sensitivity analyses.
- Contribute to statistical study design, covariate and confounder selection, and preparation of manuscripts.
- Second-author manuscript on alcohol use disorder and associated early-life exposures is currently in final

preparation.

- Conduct computational research in spatial transcriptomics, developing geometry-aware statistical representations to characterize spatial gene-expression patterns and tissue-level heterogeneity in Alzheimer's disease.

Researcher: Air Pollution and Respiratory Health 2019–2021
National Institute of Fundamental Studies, Sri Lanka

- Investigated associations between ambient particulate matter ($PM_{2.5}$ and PM_{10}) and respiratory hospitalizations using generalized additive models with quasi-Poisson regression.
- Evaluated lagged exposure effects and conducted age and sex-stratified analyses to identify populations at elevated risk.
- Led the statistical analysis and published a **first-author peer-reviewed paper**.

Researcher : Parental BMI Trajectories and Offspring Asthma 2020–2021
National Institute of Fundamental Studies, Sri Lanka

- Applied group-based trajectory modeling to Tasmanian Longitudinal Health Study data and used multinomial logistic regression to assess associations with offspring asthma outcomes.
- Identified exposure-specific potential confounders and developed adjusted statistical models.
- Co-authored a **peer-reviewed publication**.

Publications and Preprints

1. **Priyankara, S.**, Ellingson, L. (2026). FUSED: A Functional Representation for Joint Structural and Elemental Analysis of Protein Ligand Binding Sites. bioRxiv preprint; manuscript submitted to the *Journal of Chemical Information and Modeling*. **First author.**
2. **Priyankara, S.**, et al. (2021). Ambient $PM_{2.5}$ and PM_{10} Exposure and Respiratory Disease Hospitalization in Kandy, Sri Lanka. *International Journal of Environmental Research and Public Health*. **First author.**
3. Bowatte, G., Bui, D. S., **Priyankara, S.**, et al. (2022). Parental preconception BMI trajectories from childhood to adolescence and asthma in the future offspring. *Journal of Allergy and Clinical Immunology*.

Manuscripts in Preparation

1. **Priyankara, S.**, Ellingson, L. Element-Specific Axis Distance Density (ESADD) Representations of Protein Binding Sites. *Manuscript in preparation.*
2. Manuscript on alcohol use disorder and associated early-life exposures. *Manuscript in final preparation; second author.*

Conference Presentations

1. **Priyankara, S.**, Ellingson, L. (2024). *Analysis of Covariance Matrix Distances in Protein Binding Sites.* Conference of Texas Statisticians.
2. **Priyankara, S.**, et al. (2021). *PM2.5 Exposure and Respiratory Disease Hospitalization in Sri Lanka.* Lung Science Conference – European Respiratory Society.
3. Priyankara, S., Abeysundara, S.P. (2020). *Early Recognition of Premature Birth.* International Conf. on Environmental and Medical Statistics.

Professional and Teaching Experience

Graduate Instructor of Record 2022–Present
Department of Mathematics & Statistics, Texas Tech University

Graduate Teaching Assistant 2021–2022
Department of Mathematics & Statistics, Texas Tech University

Research Assistant 2019–2021
National Institute of Fundamental Studies, Sri Lanka

Technical Skills

Languages:	R, Python, STATA, MATLAB, C
Statistical Methods:	Functional data analysis, object-oriented data analysis, regression modeling (linear, logistic, multinomial), longitudinal data analysis, generalized additive models (GAM), multivariate statistics, time series analysis, survival analysis
Computational Methods:	Statistical modeling of protein structures, geometry-aware modeling, Riemannian geometry, high-dimensional data analysis,machine learning
Domain:	Ligand-protein binding prediction, spatial transcriptomics,epidemiology

Professional Memberships and Service

- Member, American Statistical Association.
- President, Statistical Circle, University of Peradeniya, 2017–2018.
- Volunteer, Refugee Services, Lubbock, Texas, 2023.
- Committee Member, Sri Lankan Students Association, Texas Tech University, 2022–2023

References

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Relationship: Research Collaborator

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Relationship: Ph.D. Committee Member